

Executive Summary

This project aims to develop an autonomous robot that sorts coloured balls, representing different plastic types, into their correct categories within a defined arena. Accordingly, this report presents the design team's understanding of the problem the project addresses, explains the chosen design in detail, and outlines associated risks and their management strategies. It also includes a detailed project plan and Gantt chart to support the successful completion of the robot.

The proposed design for the project is a Wall-E inspired robot based on a PLA chassis. Key design features include two front drive wheels and a rear ball caster wheel for mobility as well as an Arduino Uno microcontroller for autonomous control. It also features multiple batteries for power and several sensors to detect balls and bases and obstacles. To collect balls, the robot moves towards a target ball and aligns it with its clamps. These then close and the robot returns to base to deposit the ball. During development, emphasis was placed on making the design simple, reliable, affordable, and easy to modify. A subsystem breakdown was used, which allowed each part of the robot to be developed simultaneously and adjusted easily.

However, the design is limited in scope and application. It is highly specialised for ball collection within a controlled arena and cannot be applied directly to plastic waste sorting without significant modification. Additionally, the robot's batteries do not provide sufficient capacity for continuous operation over several hours, restricting its practicality in real-world applications.

Overall, it is anticipated that the project will be completed by the deadline of 23 April 2026 and that all required objectives will be achieved.

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1 Introduction

The purpose of this report is to present and explain the proposed design of the prototype waste sorting robot, as well as identify key project risks and outline the project timeline.

On university campuses and in public environments, plastic waste is often sorted improperly. When different types of plastic are incorrectly sorted together, they cannot be recycled and must instead be sent to landfill. As the volume of plastic waste produced worldwide continues to rise, manual sorting becomes increasingly unsuitable to address this problem [1]. Autonomous waste sorting technologies offer a solution by providing faster, more accurate, and cost-effective sorting [1].

The design problem of this project is therefore to develop a prototype of an autonomous robot that navigates and accurately sorts plastic waste, represented by coloured balls, into PET, HDPE, and PP categories. The robot must identify and avoid non-recyclable materials, represented by yellow balls. It must also be self-contained with maximum dimensions of 200 mm $\times$ 200 mm $\times$ 200 mm and use no more than \$50 in additional materials beyond the provided kit. The project must be completed and ready to test by 23 April 2026.

The format of this report is as follows. First, the Design Concept section provides a system overview, summarises how each component works and interfaces with others to achieve the design objectives, and outlines the team's consideration of alternative designs. Second, the Relevant Knowledge section provides in-depth technical explanations of how each subsystem functions and interacts to form the complete system. Third, the Risk Management section identifies risks and outlines mitigation strategies and backup plans. Fourth, the Project Plan section presents a Gantt chart showing task breakdowns, deadlines, and task assignments, with justification for the schedule. Finally, the Conclusion summarises the key findings and outcomes of the report.

2 Design Concept

2.1 System Overview

The proposed design is an autonomous, Wall-E-inspired robot that sorts coloured balls representing different types of plastic waste into their correct categories. The overall design of the robot is shown in Figure 1.

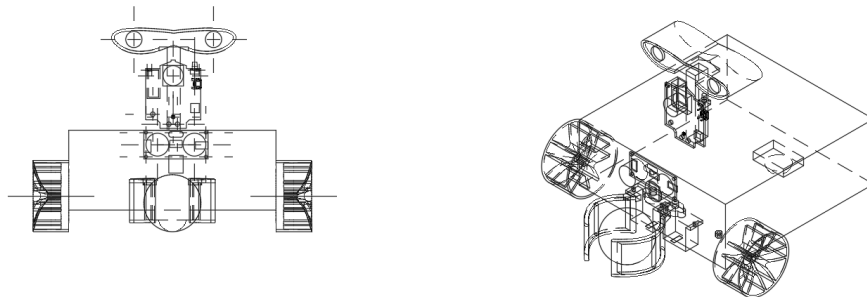


Figure 1: Technical drawing of the overall robot design

To achieve the project objectives, the robot has been divided into four distinct subsystems. This modular approach enables independent development of each subsystem and allows modifications without completely redesigning the robot. The four subsystems are:

- Control Subsystem
- Vision and Detection Subsystem
- Chassis and Navigation Subsystem
- Ball Handling Subsystem

Figure 2 illustrates the basic interactions between subsystems. Further detailed explanations of these interactions are provided throughout the Relevant Knowledge section.

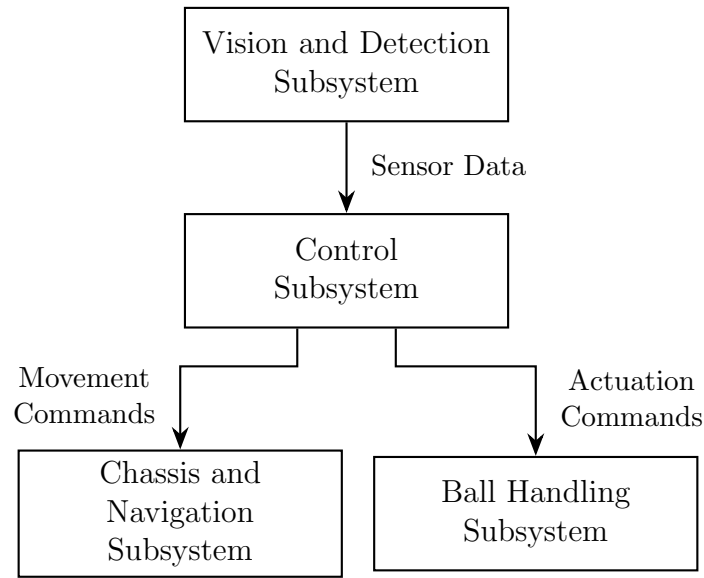


Figure 2: Block diagram of the subsystems and their interactions

2.2 Subsystem Specifications

2.2.1 Control Subsystem

The Control Subsystem is responsible for managing the operation of the entire robot. It receives sensor data from the Vision and Detection Subsystem, which it uses to send movement commands to the Chassis and Navigation Subsystem and actuation commands to the Ball Handling Subsystem. This coordination thereby enables the robot to autonomously navigate, collect, and sort the balls.

The Control Subsystem consists of an Arduino Uno microcontroller [2]. This component was selected for its affordability and simple interfacing via input-output pins. These pins enable easy connection and control of all motors, servos, and sensors across the system.

2.2.2 Vision and Detection Subsystem

The Vision and Detection Subsystem is responsible for gathering information about the robot's surroundings. It identifies coloured balls and base, measures distance to obstacles, and detects when a

ball enters the clamp, sending this data to the Control Subsystem. This sensory data enables the robot to navigate towards target balls or base, avoid obstacles, and actuate the clamp accurately.

The Vision and Detection Subsystem consists of a Pixy2 camera, an HC-SR04 ultrasonic sensor, and an FC-51 infrared sensor. The Pixy2 was selected for its built-in colour tracking and simple training process. It detects different coloured balls and bases, sending position data to the Arduino which commands the motors to move towards target balls or return to base. The ultrasonic sensor measures distance to obstacles, sending this data to the Arduino which commands the motors to brake when obstacles are too close. The IR sensor detects balls in the clamp zone, sending this data to the Arduino which commands the servos to open or close the clamp.

2.2.3 Chassis and Navigation Subsystem

The Chassis and Navigation Subsystem is responsible for providing structural support for all components and physically moving the robot throughout the arena. It receives movement commands from the Control Subsystem to control motors, enabling the robot to navigate towards target balls, return to base, and avoid obstacles.

The Chassis and Navigation Subsystem consists of a 3D-printed PLA chassis, an MKRVRS TC78H660 motor driver, two DC motors, two front wheels, and one rear ball caster wheel. The PLA chassis was selected for its strength and light weight, which aids manoeuvrability whilst supporting all components securely. The chassis has two levels to mount all components. The MKRVRS motor driver independently controls both DC motors by varying their speed and direction or applying braking. This enables the robot to move forward, move backward, turn, and stop precisely. The DC motors convert electrical signals from the motor driver into rotational force, which drives the front wheels to move the robot. The rear ball caster wheel provides a low-friction pivot point, enabling the robot to rotate on the spot for precise positioning during ball collection and deposit.

2.2.4 Ball Handling Subsystem

The Ball Handling Subsystem is responsible for collecting and releasing balls during operation. It receives actuation commands from the Control Subsystem to control the clamp mechanism, enabling the robot to collect balls and deposit them at base.

The Ball Handling Subsystem consists of two servo motors and 3D-printed clamp arms. The servo motors were selected for their precise angular control and simple interfacing with the Arduino. Actuation commands from the Arduino trigger the servos to rotate the clamp arms to open or close positions, creating or releasing the grip on a ball. The clamp arms were designed with a curved shape to enclose a ball securely. A technical drawing of the clamp mechanism is shown in Figure 3.

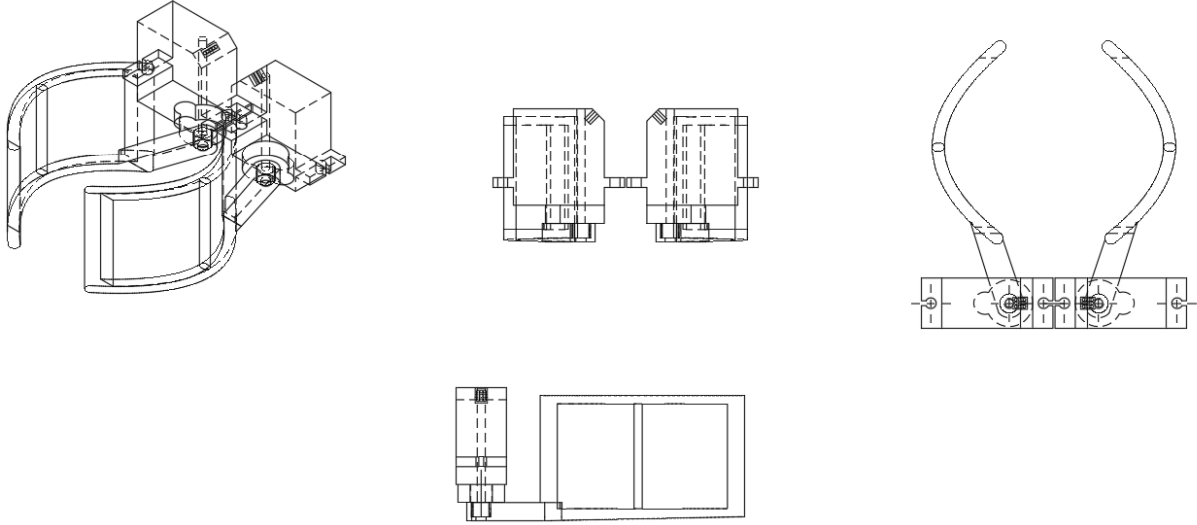


Figure 3: Technical drawing of clamp mechanism

2.3 Consideration of Alternative Designs

To select the final design for each subsystem and the overall robot, the team considered several alternative designs. A morphological chart was developed to organise and compare all the design concepts systematically. Using this approach, the team could evaluate and assess how well different concepts would work together. Based on this analysis, the most promising options were selected. Additionally, the morphological chart serves as a reference tool, allowing the team to revisit alternative solutions if a chosen approach proves unfeasible. The complete morphological chart is provided in Appendix A.

3 Relevant Knowledge

3.1 Control Subsystem

3.1.1 Power Supply

The Arduino is powered by a 9V lithium polymer battery connected through a barrel jack. The onboard voltage regulator converts the 9V input to 5V, which powers the microcontroller and provides regulated voltage to other connected components.

3.1.2 Sensor Inputs and Decision Logic

The Arduino receives sensor data from three components in the Vision and Detection Subsystem. The Pixy2 camera identifies coloured balls and bases using its built-in Colour Connected Components algorithm. Additionally, the HC-SR04 ultrasonic sensor measures distance to obstacles, and the FC-51 infrared sensor detects when a ball enters the clamp. The Arduino processes this sensor data using conditional logic in the code to determine commands. This loop continuously repeats by reading sensor inputs, evaluating conditions, and sending commands to the Chassis and Navigation Subsystem or the Ball Handling Subsystem.

3.1.3 Control Outputs

The Arduino sends commands to other subsystems using Pulse Width Modulation (PWM) signals [3]. PWM controls average voltage by adjusting the duty cycle, which is the proportion of time that a square wave signal stays high. For the MKRVRS motor driver, a higher duty cycle produces a higher average voltage, resulting in faster motor rotation. For servo motors, the pulse width determines the rotation angle, and the Arduino sends continuous signals to maintain the desired position.

3.1.4 Connection Diagram

Figure 4 and Table 1 show the pin connections between the Arduino and all the robot components.

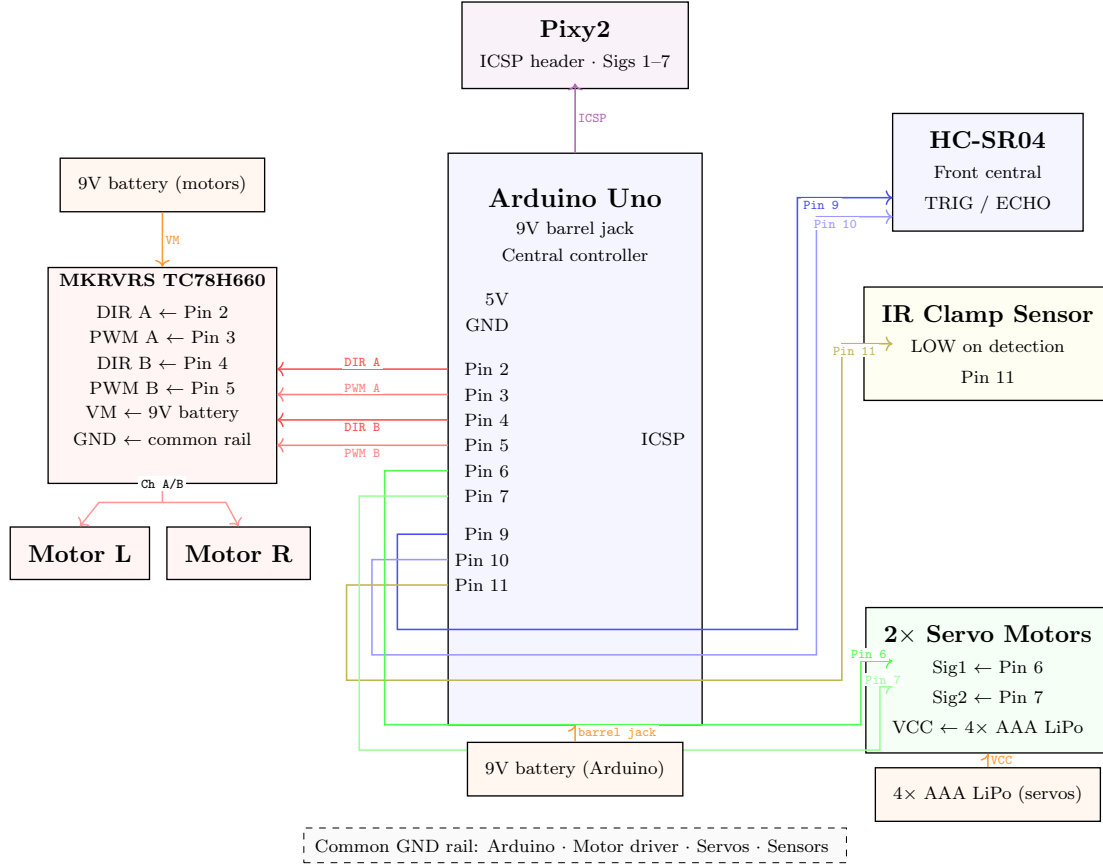


Figure 4: Diagram of Arduino Uno pin connections

Pin	Component	Function	Direction
ICSP	Pixy2	SPI data + power	Bidirectional
2	MKRVRS TC78H660	DIR A	Output
3	MKRVRS TC78H660	PWM A	Output
4	MKRVRS TC78H660	DIR B	Output
5	MKRVRS TC78H660	PWM B	Output
6	Servo 1	Signal	Output
7	Servo 2	Signal	Output
9	HC-SR04	TRIG	Output
10	HC-SR04	ECHO	Input
11	IR clamp sensor	Ball detect (LOW)	Input

Table 1: Arduino Uno pin connections

3.2 Vision and Detection Subsystem

3.2.1 Pixy2 Camera Operation

The Pixy2 detects coloured objects using trained colour signatures in Colour Connected Components mode [4]. Signatures 1 to 3 identify red, green, and blue balls respectively, while signatures 4 to 7 identify the four corner bases. The camera stores base colour signatures at startup. When an object is detected, the Pixy2 reports its position as X–Y coordinates and signature pixel size [5]. The camera communicates with the Arduino via SPI protocol through the ICSP pins [6], where the Arduino requests or receives data. The Pixy2 receives 5V power through the ICSP pins.

3.2.2 HC-SR04 Ultrasonic Sensor Operation

The HC-SR04 measures distance using 40 kHz ultrasonic sound waves [7]. The Arduino sends a trigger signal to the sensor, which causes it to emit a short ultrasonic burst. This sound wave travels forward until it hits an obstacle and reflects back. The sensor detects the returning echo and measures how long the pulse was absent. The Arduino then calculates the distance using this time measurement. The sensor receives 5V power from the Arduino’s 5V output pin.

The ultrasonic sensor can output inaccurate readings due to external noise, which creates spikes in distance measurements. To address this, a median filter is applied in the code to discard outliers from batches of five measurements and output the median value. Figure 5 shows the median filter implementation.

```
int medianFilter() {
    int readings[5];

    // Take 5 raw readings with short gap between each
    for (int i = 0; i < 5; i++) {
        readings[i] = sonar.ping_cm();
        delay(10);
    }

    // Bubble sort readings ascending
    for (int i = 0; i < 4; i++) {
        for (int j = 0; j < 4 - i; j++) {
            if (readings[j] > readings[j + 1]) {
                int temp = readings[j];
                readings[j] = readings[j + 1];
                readings[j + 1] = temp;
            }
        }
    }

    return readings[2]; // middle value = median
}
```

Figure 5: Code excerpt of median filter implementation using bubble sort

Multiple robots operating in the arena can also cause interference, as ultrasonic bursts from other sensors may trigger false short-distance readings on the echo pin. To reduce the likelihood of this occurring, a time-jitter technique is implemented in the code, which randomises the trigger timing. Figure 6 shows this implementation.

```

void loop() {

    // Read distance via median filter
    int distance = medianFilter();

    if (distance > 0 && distance < OBSTACLE_THRESHOLD) {
        stopMotors();
    }

    // Random delay - breaks synchronisation with other robots'
    // ultrasonic sensors to prevent crosstalk false positives
    delay(random(50, 150));
}

```

Figure 6: Code excerpt of the time jitter technique

3.2.3 FC-51 Infrared Sensor Operation

The FC-51 IR sensor detects nearby objects using infrared light reflection. The sensor is positioned between the clamp arms and continuously emits infrared light. When a ball enters the clamp zone, the infrared light reflects back to the sensor, causing it to output a LOW signal. When no ball is present, the sensor outputs HIGH. The Arduino reads this digital signal on a designated pin and triggers clamp closure when a ball is detected. The sensor receives 5V power from the Arduino.

3.2.4 Sensor Mounting

The Pixy2 is mounted on the top chassis level with an adjustable angle mount for fine-tuning, allowing it to view both base colours and the clamp mechanism. The HC-SR04 is mounted front-central for forward obstacle detection. The FC-51 IR sensor is integrated into the clamp mechanism for ball detection.

3.3 Chassis and Navigation Subsystem

3.3.1 Chassis Design

The chassis is 3D-printed using PLA material with 50% infill, which provides strength whilst remaining lightweight [8]. The chassis features a two-level design. The bottom level houses the motors and batteries, whilst the top level holds the Arduino and Pixy2. This arrangement places heavier components low for improved stability. The chassis base measures 146 mm × 145 mm, which fits within the robot’s size constraints. Precision mounting points are integrated into the design to secure all components.

3.3.2 Power Supply

A dedicated 9V lithium polymer battery powers the motors via the motor driver. This battery is separate from the Arduino power supply to prevent voltage drops during motor operation. The motors can draw up to 1.1 A each, necessitating a dedicated high-current power source.

3.3.3 MKRVRS Motor Driver Operation

The motor driver receives PWM signals from the Arduino to control motor speed and direction. An H-bridge is an electronic circuit that allows voltage to be applied across a load in either direction, enabling motor direction control [9]. The MKRVRS uses an H-bridge circuit with four transistor switches to allow current flow reversal. Figure 7 shows the H-bridge circuit configuration.

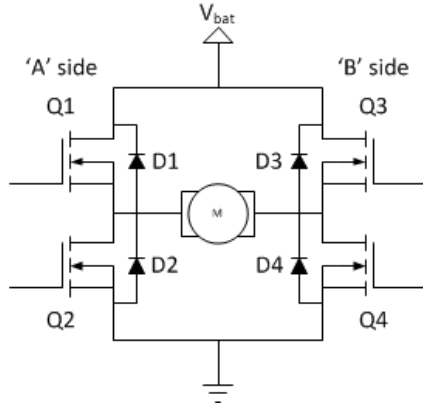


Figure 7: Diagram of H-bridge circuit

The PWM duty cycle controls the average voltage delivered to the motors. A higher duty cycle produces higher average voltage and results in faster motor rotation. The motor driver independently controls both DC motors, enabling differential steering.

3.3.4 DC Motor Operation

The DC motors convert electrical signals from the motor driver into rotational force, which drives the front wheels through attached axles. The motors operate within a 3 to 9 V range, where the applied voltage determines the rotation speed. The motors draw current proportional to the mechanical load they experience.

Differential steering enables directional control [10]. For forward movement, both motors run at the same speed and in the same direction. For turning, the motors run at different speeds in the same direction, where the slower inner wheel and faster outer wheel create an arc. For rotation on the spot, the motors run at the same speed in opposite directions, causing the robot to pivot around the rear caster wheel. The rear ball caster wheel provides a low-friction pivot point, enabling precise positioning.

3.3.5 Movement Algorithm

The movement algorithm is outlined in the Subsystem Interaction section.

3.4 Ball Handling Subsystem

3.4.1 Power Supply

Two servo motors are powered by four AAA lithium polymer batteries connected in series. A separate power source is used to ensure sufficient current for servo operation without affecting the Arduino's power supply. The motors can draw up to 500 mA each during movement, necessitating dedicated power to avoid overloading the Arduino.

3.4.2 Servo Motor Operation

The servo motors control clamp arm position using PWM signals from the Arduino. The PWM pulse width determines the rotation angle. An internal potentiometer measures the shaft position, and the control circuit adjusts the motor until the measured position matches the target position indicated by the pulse width [11]. Both servos receive commands for the same angle to achieve synchronised

movement of the clamp arms. The servo response time is 0.3 seconds. When idle or holding position, each servo draws approximately 100 mA, whilst during movement the current draw increases to approximately 500 mA.

3.4.3 Clamp Design

The clamp arms are 3D-printed with a curved design to match the ball diameter. The arms rotate inward simultaneously when the servos actuate to create a grip on the ball and secure it during transport.

3.4.4 Collection and Release Sequence

During collection, the Pixy2 camera navigates the robot towards the target ball based on colour signature detection. Once nearby, the Pixy2 centres the ball in its frame and aligns it with the open clamp. The Arduino then commands the motors to move forward at low PWM until the ball enters the clamp enclosure. When the FC-51 infrared sensor detects the ball, the Arduino commands the servos to close and secure it. The infrared sensor is mounted between the clamp arms to detect ball entry. The Arduino continuously monitors the sensor state, where HIGH indicates no ball and LOW indicates a ball is present within the detection threshold. During release, the Arduino commands the servos to open the clamp arms, allowing gravity to release the ball into the base.

3.5 Subsystem Interaction

See Figure 8 for the autonomous functioning algorithm of the robot.

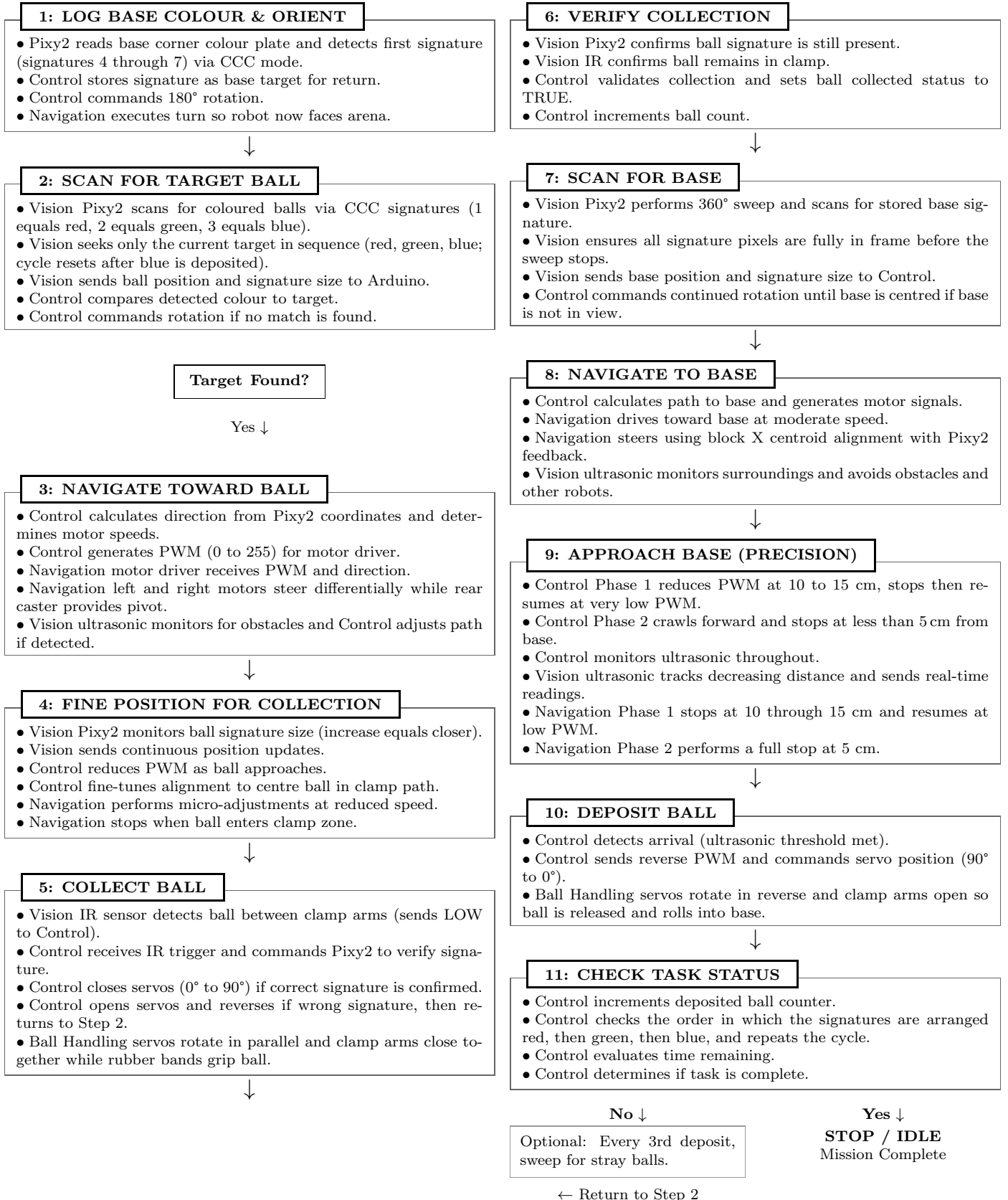


Figure 8: Flowchart of the ball collection and deposit algorithm

4 Risk Management

Within any project, both technical and non-technical risks can negatively impact the ability of a project to achieve its objectives. Risk management is therefore an important practice for identifying potential risks early and creating strategies to mitigate their likelihood and reduce their impact should they occur.

One technical risk involved in developing the robot is damage to components. Several delicate electrical components are used in this project, which can easily be overloaded or physically damaged if handled incorrectly. If components are damaged, it will cost significant time and money, as new components will need to be ordered. This may leave the robot non-functional for testing if components do not arrive in time. To mitigate this risk, datasheets will be consulted to determine the voltage and current limits of components, and all circuit designs will be peer checked to ensure their safety before building. To prevent physical damage, components will be handled carefully at all times and stored in a secure box when not in use. To reduce the impact if this risk still occurs, \$15 of the budget has been allocated to purchase replacement components if needed.

Another technical risk is subsystems not working individually or not integrating correctly. If one subsystem fails, the entire robot will become non-functional and fail testing. To minimise this risk, each subsystem will be tested individually to identify any issues, and team meetings will be held frequently to ensure that all subsystems are compatible. Trial runs will also be conducted before testing to identify and troubleshoot any remaining issues.

Battery depletion during testing poses another technical risk. If a battery dies during testing, the robot will become non-functional and fail testing, as manual intervention is not allowed. To mitigate this risk, fresh batteries will be used on testing day, and the weight of the robot will be minimised to reduce power consumption. Trial runs will also be conducted before testing to ensure the batteries have sufficient capacity to last the full duration of testing. Similarly, component failure on testing day is a significant risk. However, due to budget constraints, purchasing spare components for all parts is not possible. Therefore, thorough testing of each component will be conducted throughout the project to ensure reliability.

As mentioned previously, there are also several non-technical risks that must be considered to ensure the project's success. One non-technical risk is missing both internal and university-set deadlines. Failing to meet a deadline may cause the entire project to fall behind schedule, as tasks are interdependent. This can lead to the project not being completed on time or work being rushed, reducing the robot's quality. To minimise this risk, a detailed project plan and Gantt chart will be used to track deadlines and task assignments, with regular meetings held every few days to monitor progress. To reduce the impact if this risk still occurs, team members from less intensive or ahead-of-schedule tasks will be reassigned to provide additional support.

Another non-technical risk is the loss of a team member due to unforeseen circumstances such as illness. If a team member with key information or skills becomes unavailable, progress may be impacted. To mitigate this risk, all files will be shared via Microsoft Teams, and at least two group members will maintain a strong understanding of each subsystem, allowing work to continue if this risk occurs.

5 Project Plan

5.1 Key Points

Planning is essential for completing any project on time and to a high standard. Therefore, a comprehensive project plan has been created to guide the building of the waste sorting robot.

The team has decided to use a Gantt chart as its primary time management tool. This chart shows the project timeline, how tasks depend on each other, what resources are needed, and who is responsible for each task. The work is divided among three subgroups: Mechanical (MG), Electrical (EG), and Programming (PG). The project plan consists of three main phases: Concept Generation and Selection, Initial Design, and Final Design.

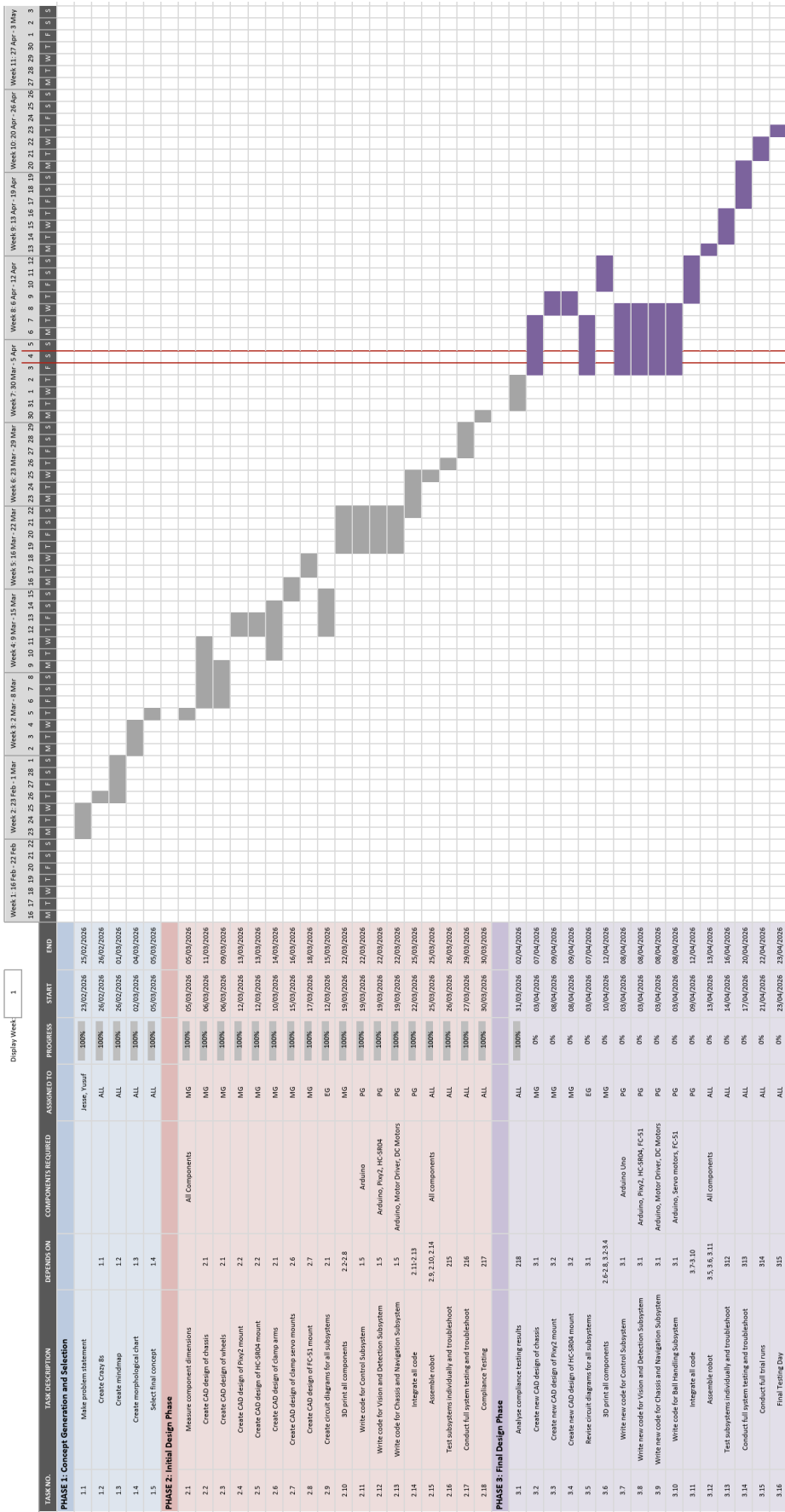
The Concept Generation and Selection Phase runs from Monday of Week 2 to Thursday of Week 3. During this phase, the team defines the problem statement, develops design concepts, and selects the best concepts.

The Initial Design Phase runs from Thursday of Week 3 to Monday of Week 7, ending with the Compliance Testing Assessment. This phase focuses on CAD design, circuit layout design, programming, and building the first iteration of the waste sorting robot. Due to time constraints, the Ball Handling Subsystem was designed but not built during this phase.

The Final Design Phase runs from Monday of Week 7 to Thursday of Week 10, the day of the Final Testing Assessment. This phase focuses on improving the first robot iteration based on testing results, adding the Ball Handling Subsystem, and completing the final build.

Throughout the entire process, the budget is strictly managed to ensure the team does not exceed the \$50 limit for additional materials beyond the provided kit. So far, only \$5 has been spent on a barrel jack connector.

5.2 Gantt Chart



6 Conclusion

Overall, the final robot design satisfies the design objectives, demonstrating the ability to autonomously identify and sort balls that represent plastic waste. This is achieved while adhering to all size, performance, and budget constraints.

The final design is divided into four subsystems: the Control Subsystem, Vision and Detection Subsystem, Chassis and Navigation Subsystem, and Ball Handling Subsystem. This structure allows for easy modification and enables each subsystem to be developed independently.

Throughout the project, the team has improved its ability to collaborate and manage the workload effectively through structured planning and task distribution. The team is satisfied with the final design and will soon begin building the prototype in preparation for final testing.

7 References

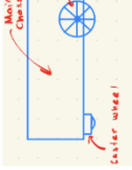
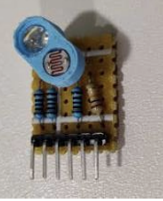
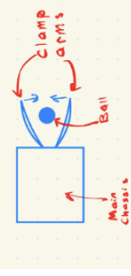
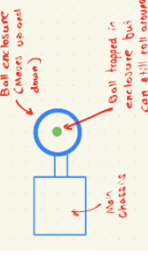
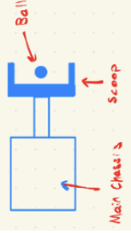
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8 Appendices

8.1 Appendix A

Design / Function	1	2	3	4
Movement	Tank tracks 	3 wheels 	2 wheels and caster 	4 wheels 
Detect ball and obstacles	Computer vision 	Parking sensors 	IR sensor 	Ultrasonic sensor 
Identify colour	Pixy2 Camera 	Colour sensor 	Spectral sensor 	LDR and RGB LEDs 
Collect and release ball	Clamp 	Bell 	Roller 	Scoop 
Navigate to correct base	Rails 	Follow lines 	Base colour detection 	Remote control 